

Course Introduction

Phase 1

- Self-Study

Phase 2

- Core Concepts
- Mini Projects

Phase 3

- Internship
- Full Projects
- Placement Assistance

Datamites Exam Portal

For Module-Level Assessments:

Link to register <https://datamites.com/exam/Registers>

PAT Service

- After 2 projects completion you are eligible for PAT service.
- Resume service.
- Complete all module level assessments.
- Complete job ready assessment.
- Mock Interview.
- Based on all above scores, if score is above 6,eligible for interview with clients.



1. 4-hr study day
2. Keep running notes
3. Multiple learning sources
4. Stick to Syllabus



Foundation Program

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Who can pursue this course?

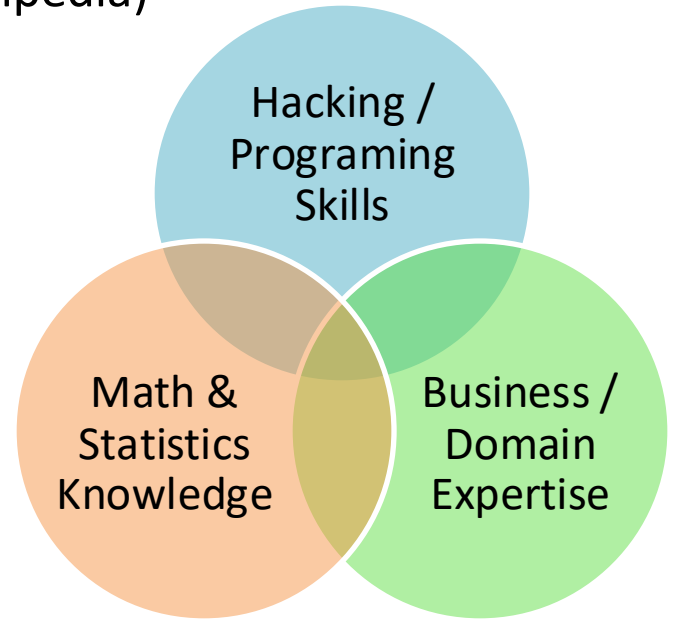
- Working Professionals looking to transition
- Data Professionals, Business Analysts, Software Engineers
- Senior Professionals, Managers
- Students, Beginners from any background
- Anyone with an interest in this field

Introduction

Some key definitions

What is Data Science?

- Data science is an interdisciplinary field that uses scientific methods, processes, algorithms and systems to extract knowledge and insights from many structured and unstructured data. (Ref: Wikipedia)
- Data science is the field of study that combines domain expertise, programming skills, and knowledge of mathematics and statistics to extract actionable insights.
- Data Science can be simply stated as “Insights from Data”



What is Analytics?

Data Analytics

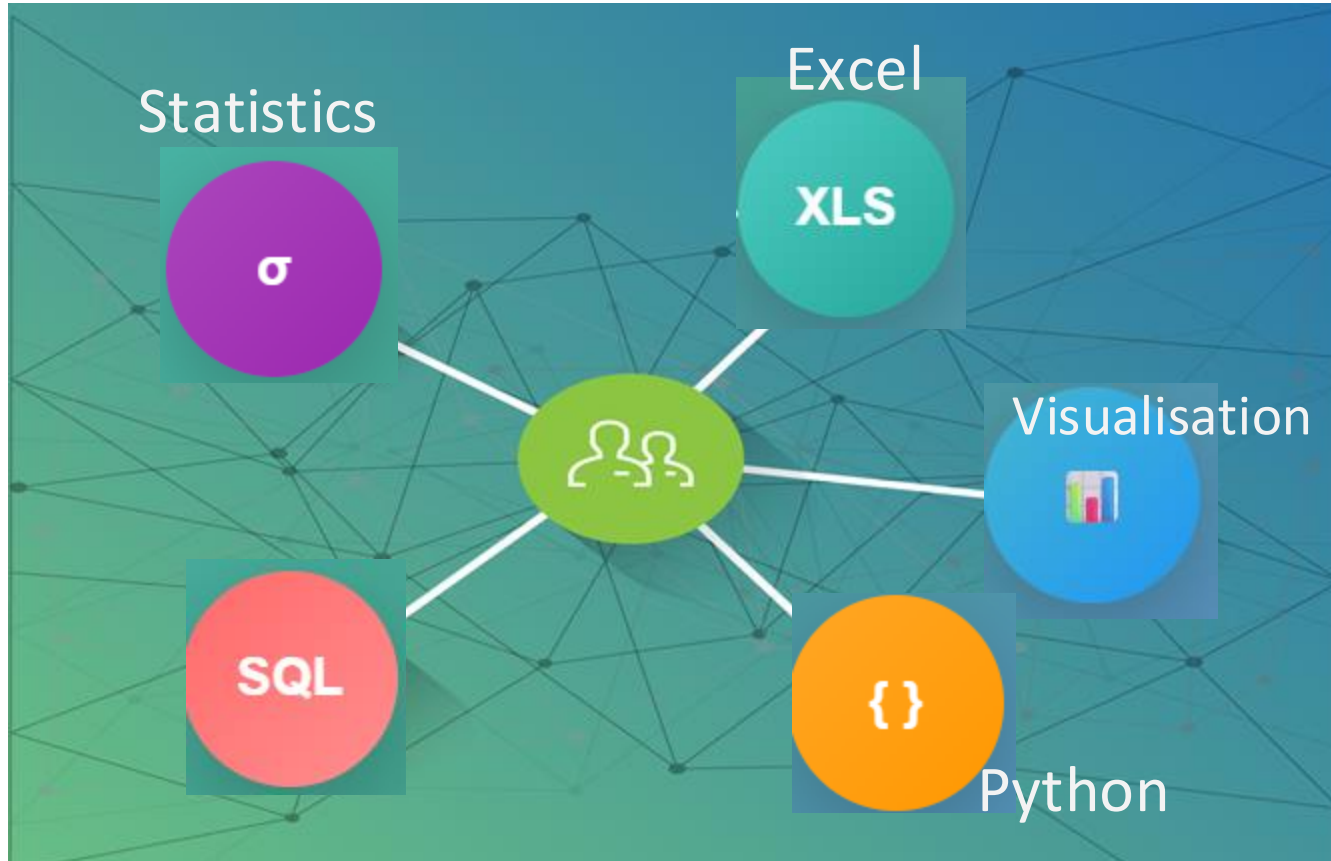
A process of examining collected data in order to find trends and draw conclusions.



Business Analytics

Harnessing data to gain insights for the benefit of the business.

Analytics in essence

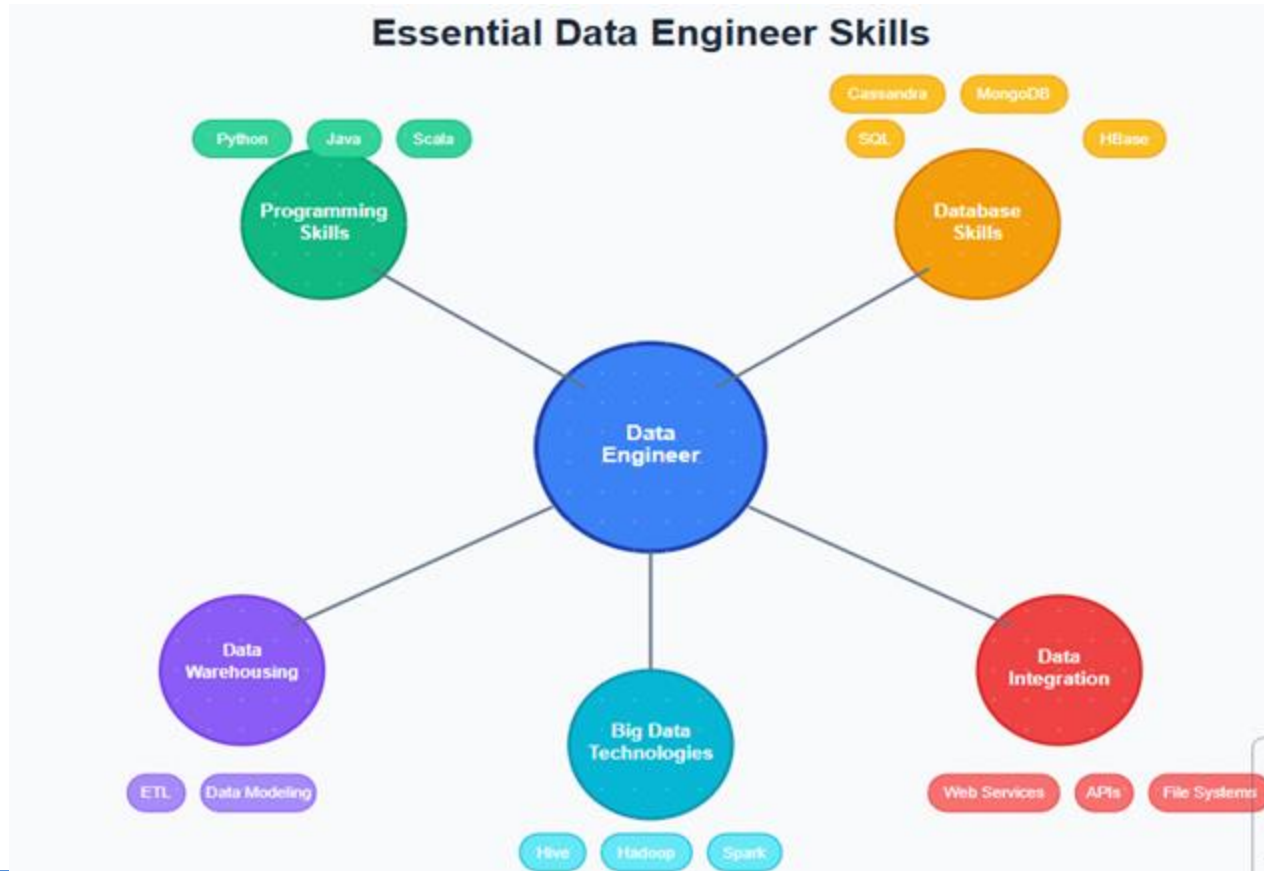


What is Data Engineering?

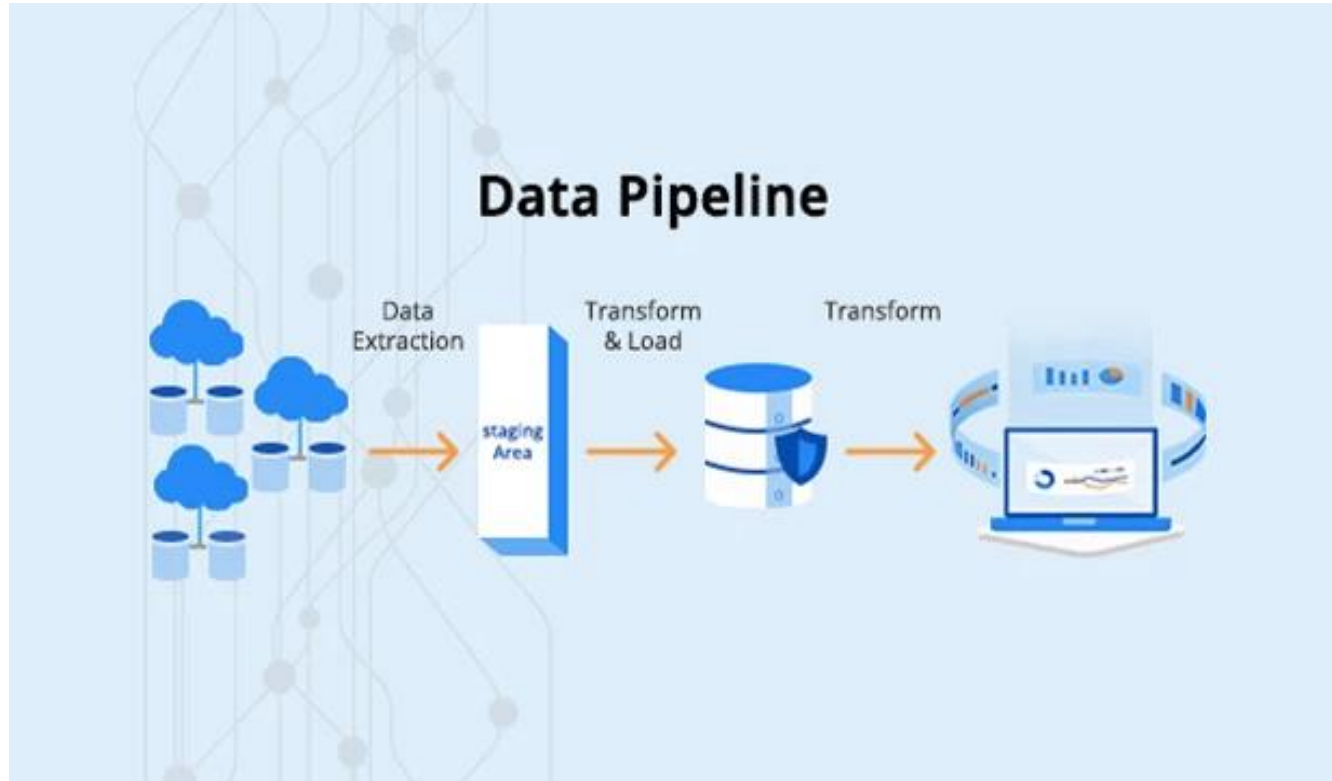
Data engineering refers to the building of systems to enable the collection and usage of data. This data is usually used to enable subsequent analysis and data science; which often involves machine learning.



Data Engineering Basics

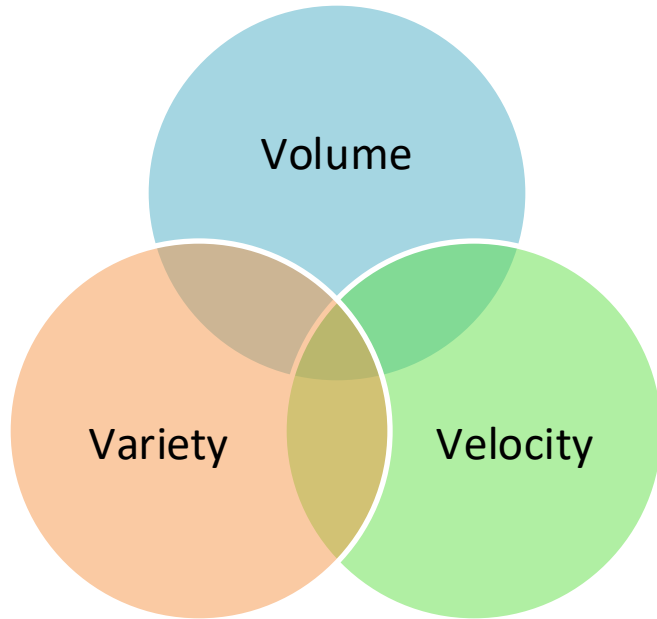


What is Big Data?



Big Data

Big Data Characteristics



Evolution



Data Science vs Business Analytics

Data Science	Business Analytics
Involves programming / coding	Minimal/No coding required.
Data Science uses both structured and unstructured data	Business Analytics uses mostly structured data
Data Science extensively involves Machine Learning and Artificial Intelligence as tools.	Business Analytics use traditional statistical and forecasting techniques.

Why SQL

- Data scientists can leverage SQL to extract, analyze, and derive meaningful insights from the relational database, informing business decisions and strategies.
- SQL is a powerful tool for efficiently handling and transforming data for analysis in real-world scenarios.

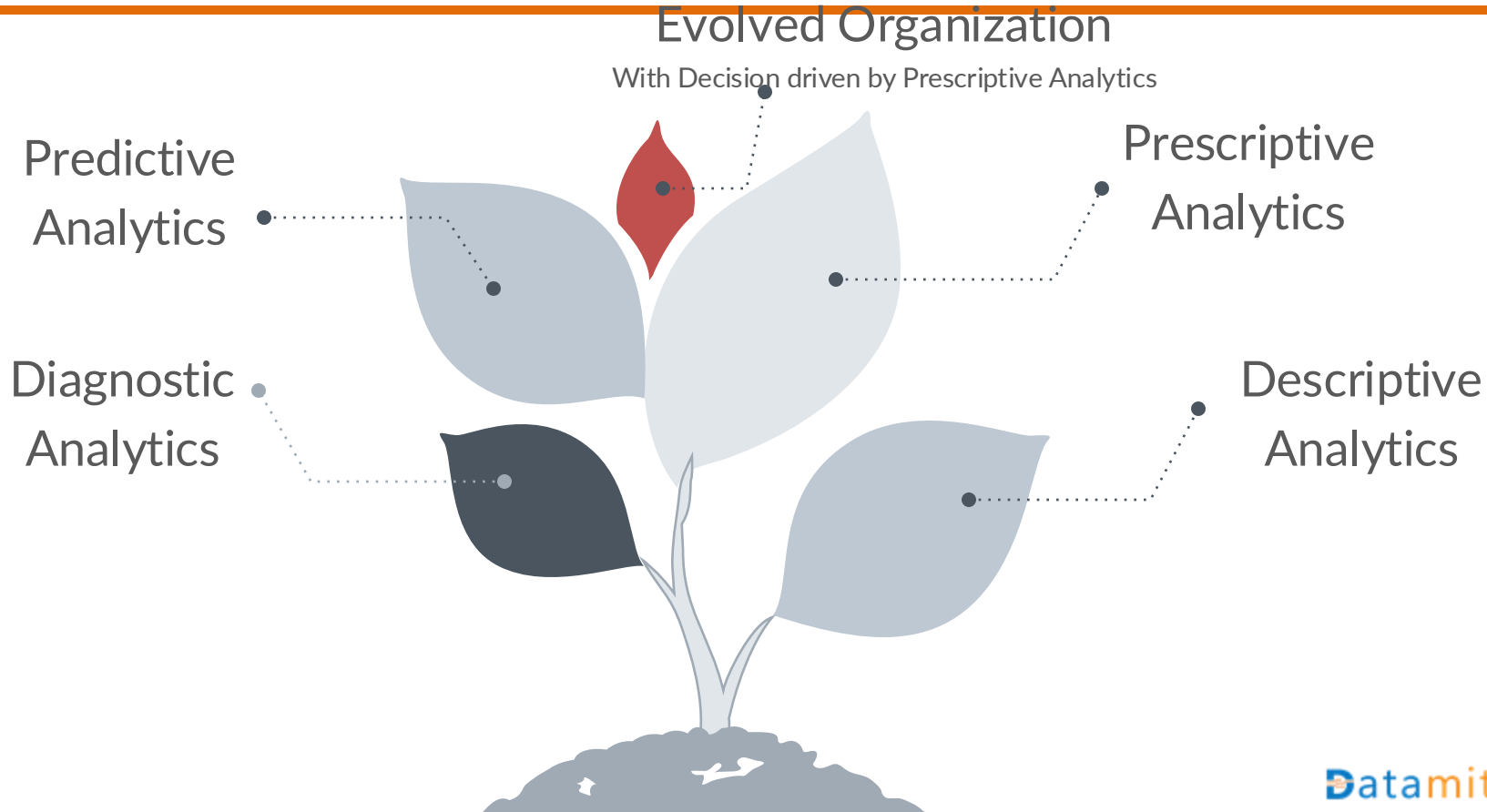


Why BI tools

- BI tools provide a user-friendly interface for data exploration, visualization, and reporting.
- Enhance the efficiency and effectiveness of data scientists in their analytical workflows.



Analytics Classification



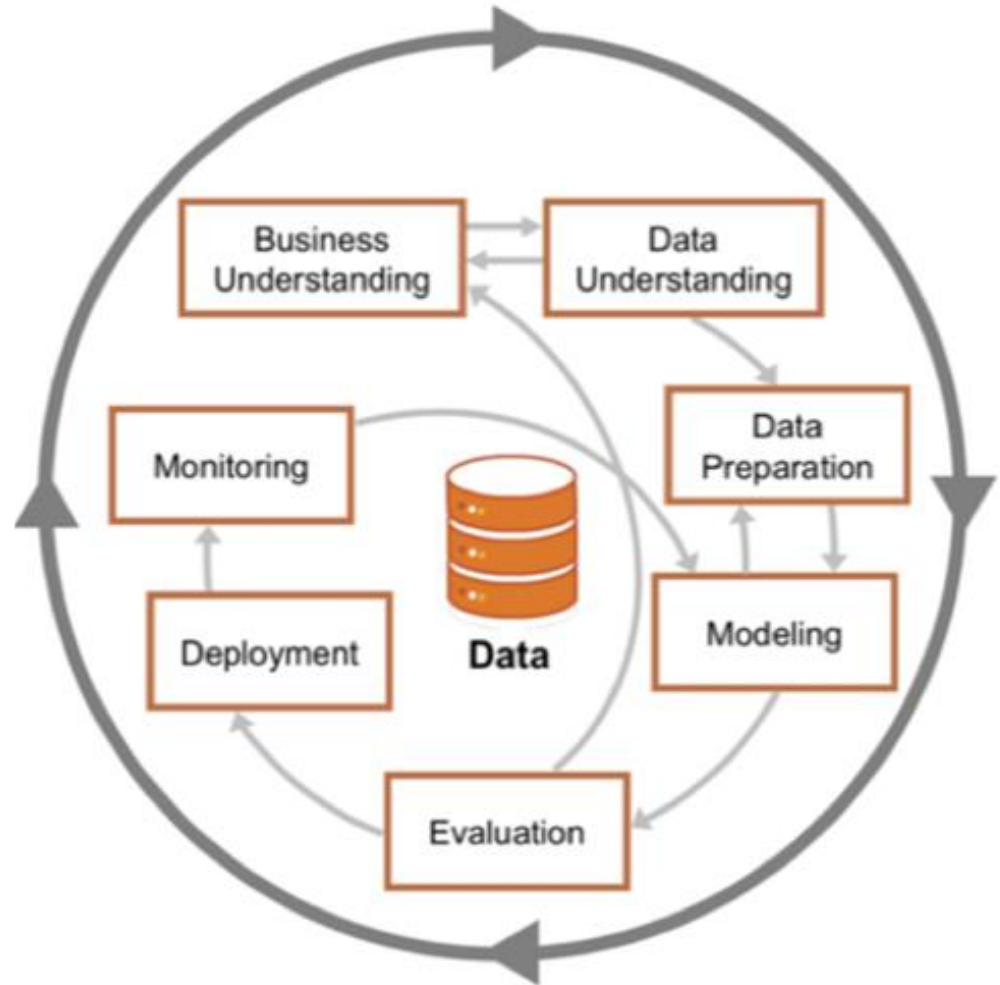
Role of Analytics in Organizations

Business analytics plays a crucial role in modern business and organizations by providing insights that can drive strategic decision-making. Here are some of the key roles of data analytics:

- 1. Identifying trends and patterns**
- 2. Improving operational efficiency**
- 3. Enhancing customer experience**
- 4. Risk management**
- 5. Business forecasting**

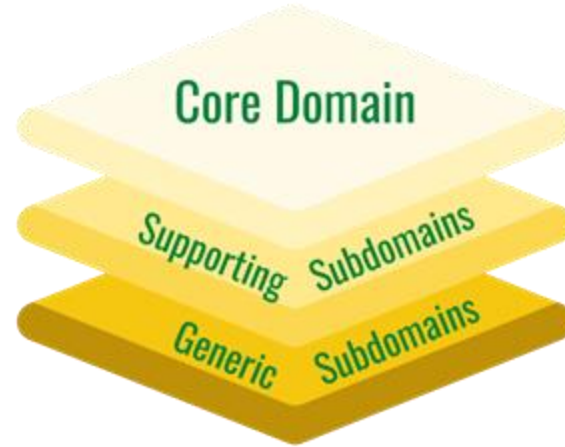
CRISP-DM Model

C*Ross Industry Standard* **P***rocess* for **D***ata* **M***ining* is a process model with different phases that naturally describes the data mining life cycle.



Domain Analysis

- It is necessary to understand the domain prior to proceeding.
- This helps to gain a business perspective on the analysis which follows.



Exploratory Data Analysis

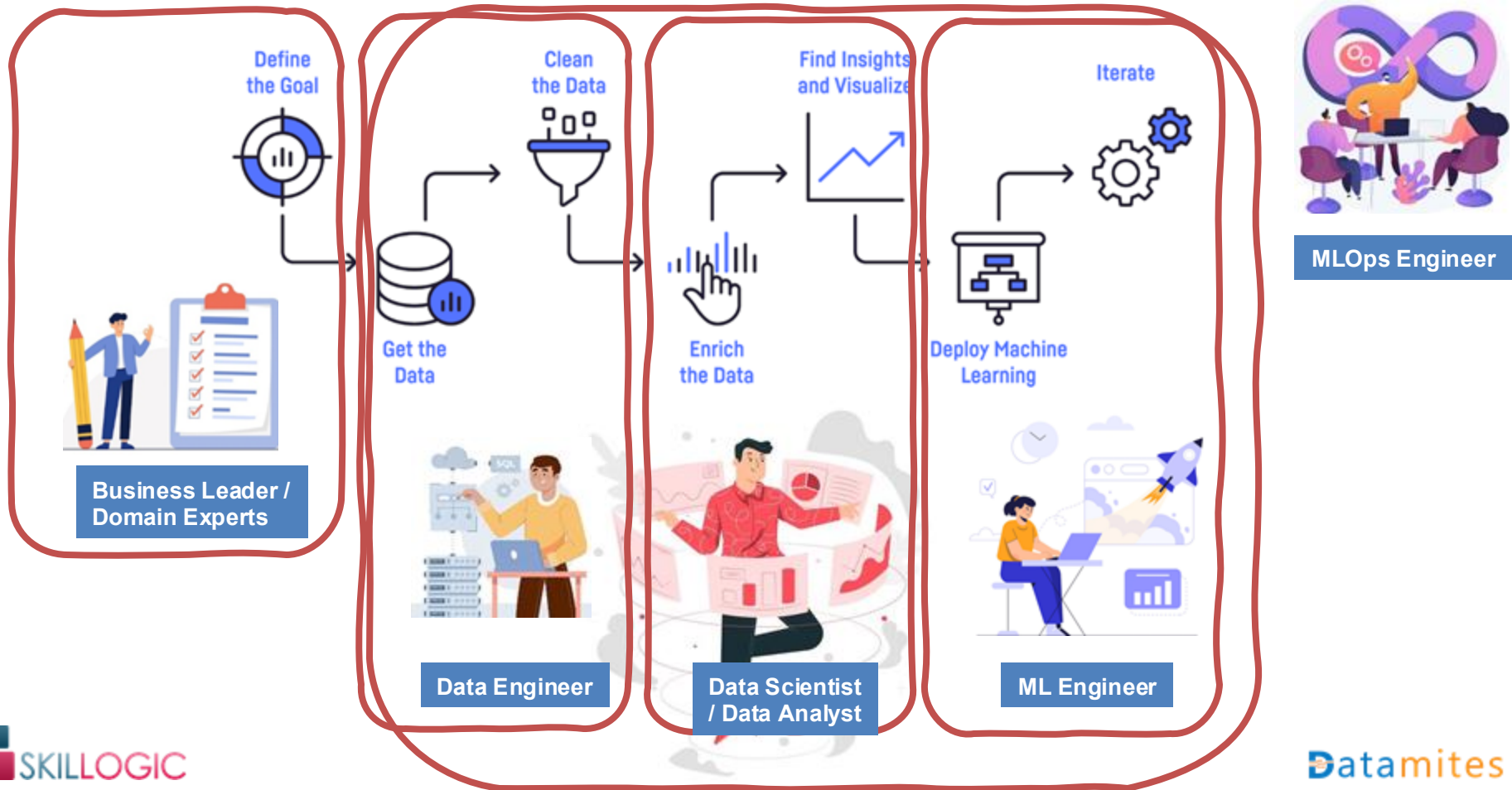
- EDA allows us to arrive at patterns and relationships within the data.
- This includes the use of visual tools, graphical analysis.



Data Preparation

- Data preparation is also known as data preprocessing.
- This is an essential stage of data science.
- Once the data is prepared, we may use it to perform predictive analytics.





Machine Learning Introduction

What is Machine Learning?

Machine learning is a field of computer science that gives computers the ability to learn without being explicitly programmed.

Areas of Artificial Intelligence



Computer Vision



Natural Language Processing



Machine Learning & Deep Learning



Decision Making



Robotics

Machine Learning Use Case Demo



Amazon Return Policy Exploitation

A Machine Learning Case Study - **Demo**

Types of Machine Learning

- Supervised machine learning
- Unsupervised machine learning
- Reinforcement machine learning

Supervised machine learning

- The most popular machine learning method.
- Uses labelled (past) data to learn and predict future events.
- The model can be evaluated and fine-tuned to attain accurate predictions.

Supervised learning

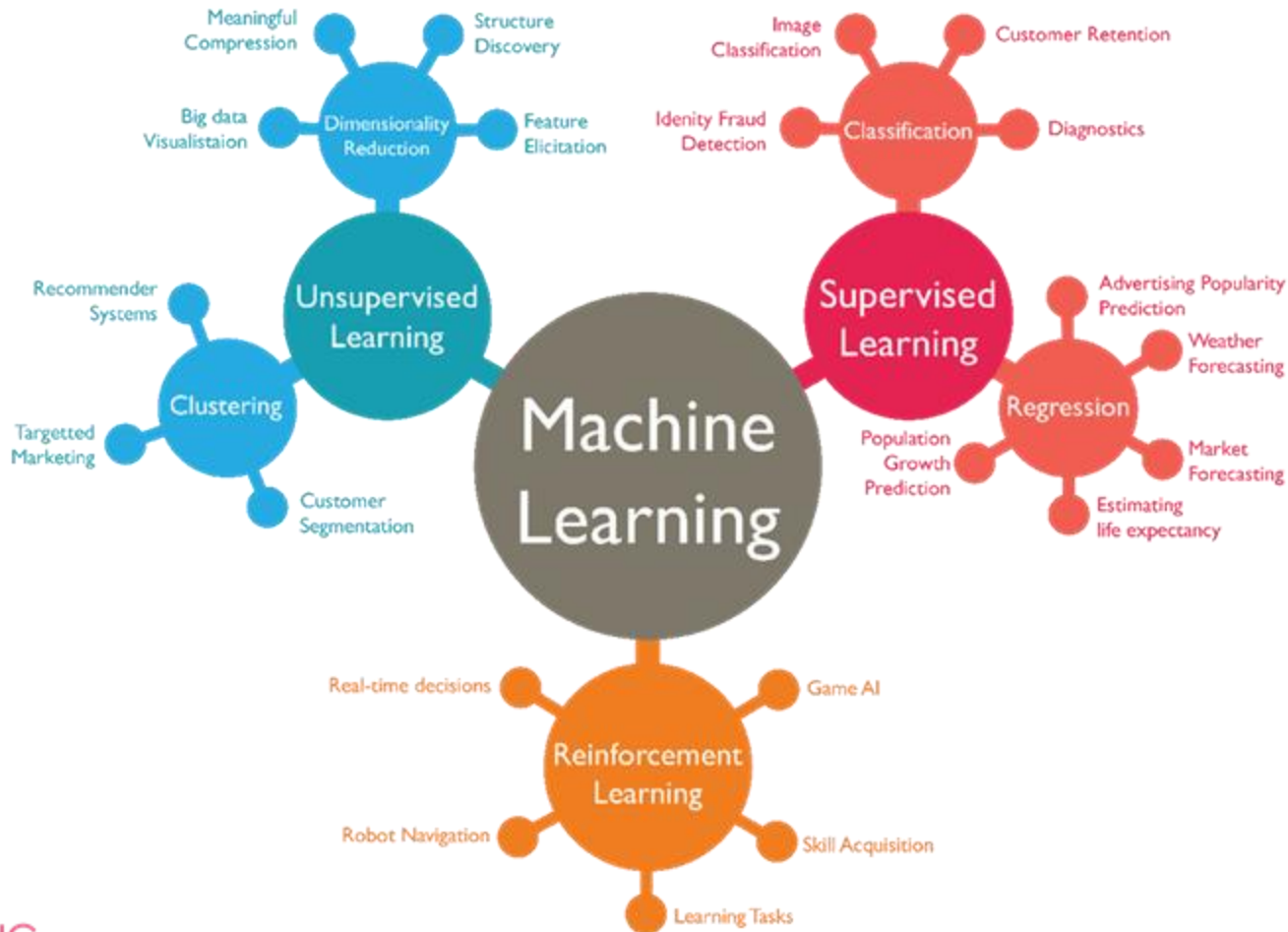
Hands on

Unsupervised machine learning

- Used when the past data used to train is neither classified nor labeled.
- The system doesn't figure out the right output, but it explores the data and can draw inferences from datasets to describe hidden structures from unlabeled data.

Reinforcement machine learning

- Learning by interacting with its environment by producing actions and discovers errors or rewards.
- This method allows machines and software agents to automatically determine the ideal behavior within a specific context in order to maximize its performance.
- Simple reward feedback is required for the agent to learn which action is best; this is known as the reinforcement signal.



Machine Learning

Machine Learning Algorithms

	<u>Unsupervised</u>	<u>Supervised</u>
<u>Continuous</u>	• Clustering & Dimensionality Reduction ○ SVD ○ PCA ○ K-means	• Regression ○ Linear ○ Polynomial • Decision Trees • Random Forests
<u>Categorical</u>	• Association Analysis ○ Apriori ○ FP-Growth • Hidden Markov Model	• Classification ○ KNN ○ Trees ○ Logistic Regression ○ Naive-Bayes

Data Science and Related Fields

Introduction to AI

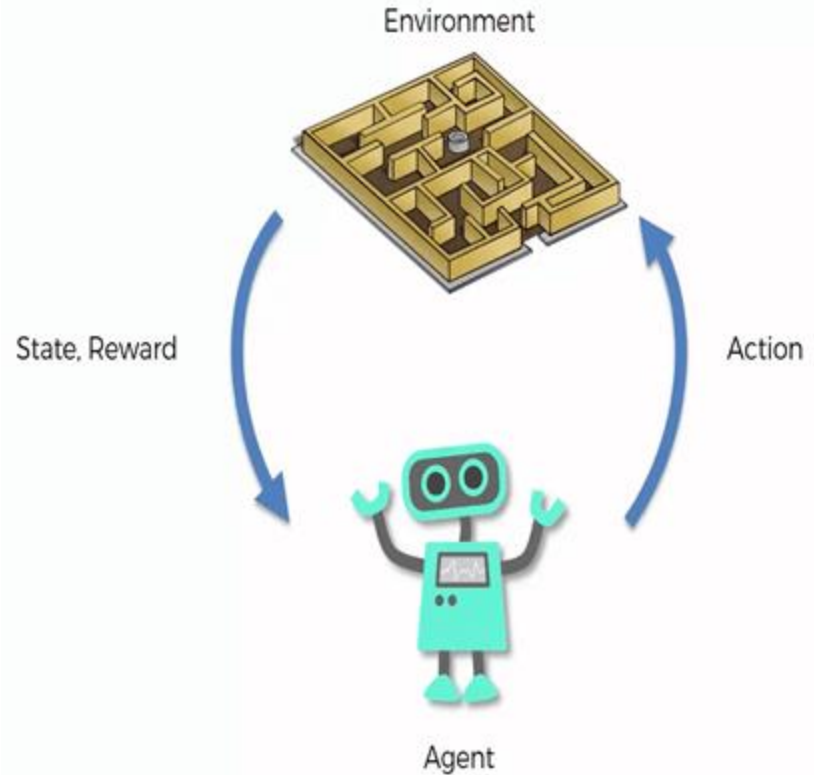
Artificial Intelligence is the simulation of human intelligence in machines that are programmed to think and learn like humans.



Reinforcement Learning

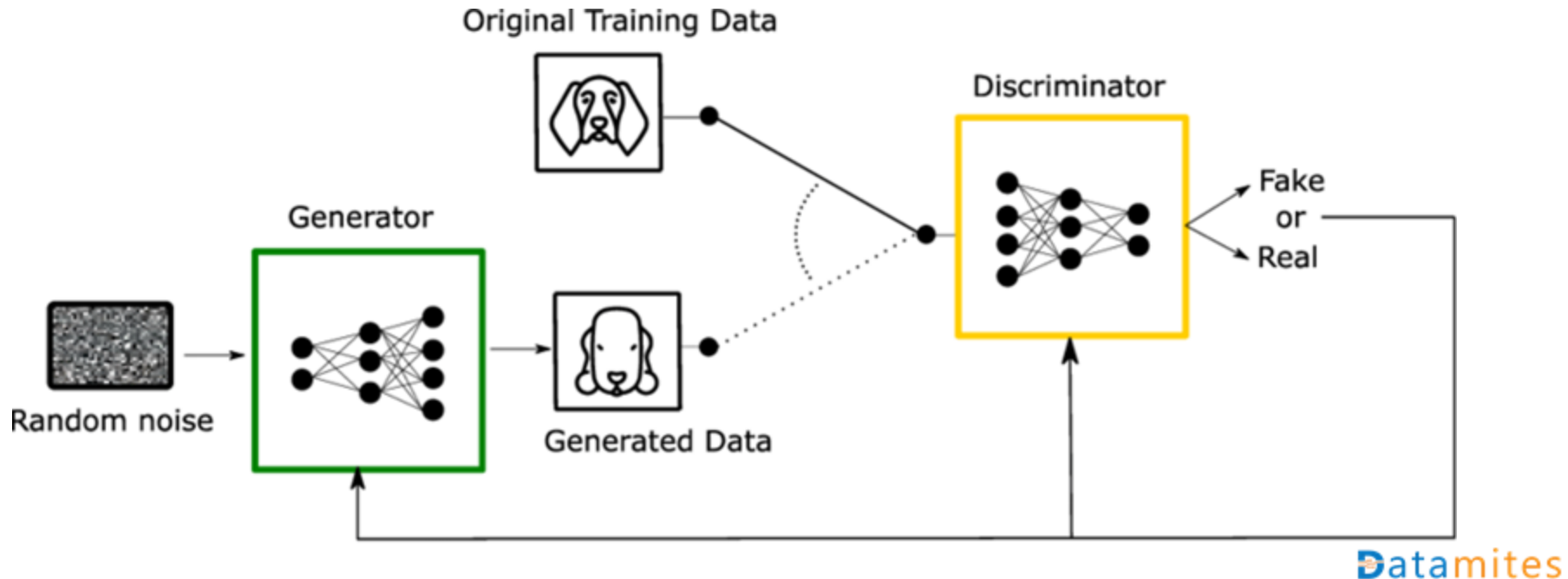
Reinforcement Learning is a type of machine learning where an agent learns to make decisions by taking actions in an environment to maximize cumulative rewards.

Applications: Gaming, robotics, and autonomous systems



GAN

GANs are a class of artificial intelligence algorithms used in unsupervised machine learning for generating synthetic data. Applications: Image generation, style transfer, and Data Augmentation



Generative Passive Model

Class of generative models that create new data instances without the need for explicit training.

Techniques: Rule-based systems, procedural generation, and evolutionary algorithms.

Applications: Content generation, Level design, and Creative arts



Data Science Industry Applications, Workflow and Job Roles

Industry Applications

Health Care

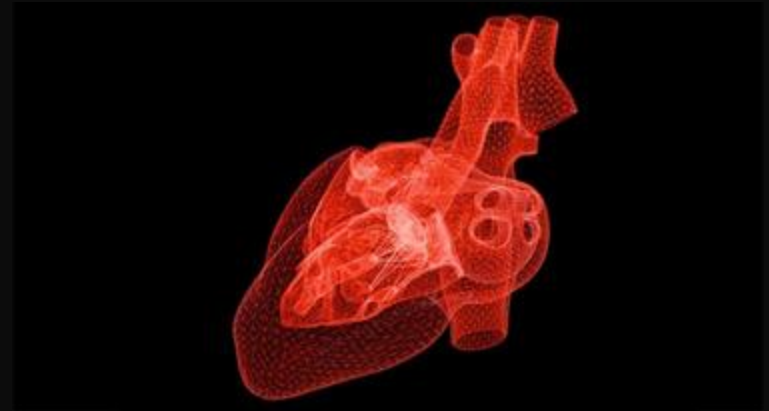
University of Oxford researchers created an AI system that scanned routine medical data to predict risk of cardiac events in the next 10 years.

<https://www.theguardian.com/society/2023/nov/13/ai-could-predict-heart-attack-risk-up-to-10-years-in-the-future-finds-oxford-study>

<https://www.ox.ac.uk/news/2023-11-13-ai-tool-could-help-thousands-avoid-fatal-heart-attacks>



AI Predict Heart Attack Better Than Doctors



AI will fight challenging diseases



We'll Soon Trust AI
More Than Doctors

From Diagnosis to Treatment

Retail

Machine Learning in Retail

what to stock, how much to buy, what products to suggest to repeat customers. But doing more with that data using machine learning is just what retailers need to really succeed in the current market.

<http://www.mckinsey.com/business-functions/mckinsey-analytics/our-insights/the-age-of-analytics-competing-in-a-data-driven-world>

<https://www.forbes.com/sites/garydrenik/2023/08/22/retail-redefined-ais-affinity-for-personalized-engagement/?sh=6065b4537a3a>



#130449165

AI in Banking



- Fraud Transactions Detection
- Predicting risk of asset class based on future context models.
- More personalized and faster customer experiences

Financial Services - American Express



- American Express processes \$1 trillion in transaction and has 110 million AmEx cards in operation.
- They rely heavily on data analytics and machine learning algorithms to help detect fraud in near real time, therefore saving millions in losses.
- Additionally, AmEx is leveraging its data flows to develop apps that can connect a cardholder with products or services and special offers.
- They are also giving merchants online business trend analysis and industry peer benchmarking.

Manufacturing - Volvo



- Cars are increasingly connected and generate data that can be used in a number of ways.
- Volvo uses data to help predict when parts would fail or when vehicles need servicing, uphold its impressive safety record by monitoring vehicle performance during hazardous situations and to improve driver and passenger convenience.
- Volvo is also conducting its own research and development on autonomous vehicles.

Media - Netflix

- Big data analytics is helping Netflix predict what its customers will enjoy watching.
- They are also increasingly a content creator, not just a distributor, and use data to drive what content it will invest in creating.
- Due to the confidence they have in the data findings, they are willing to buck convention and commission multiple seasons of a new show rather than just a pilot episode.



Retail - Burberry



- When you first think of Burberry, you likely consider its luxury fashion and not first consider them a digital business.
- However, they have been busy reinventing themselves and use big data and AI to combat counterfeit products and improve sales and customer relationships.
- The company's strategy for increasing sales is to nurture deep, personal connections with its customers.
- As part of that, they have reward and loyalty programs that create data to help them personalize the shopping experience for each customer

Service - Microsoft



- Central to everything Microsoft does is leveraging smart machines.
- Microsoft has Cortana, a virtual assistant; chatbots that run Skype and answer customer service queries or deliver info such as weather or travel updates and the company has rolled out intelligent features within its Office enterprise.
- Other companies can use the Microsoft AI Platform to create their own intelligent tools.
- In the future, Microsoft wants to see intelligent machines with generalized AI capabilities that allow them to complete any task.

Social Media - Facebook

- Deep learning is helping Facebook draw value from a larger portion of its unstructured datasets created by almost 2 billion people updating their statuses 293,000 times per minute.
- Most of its deep learning technology is built on the Torch platform that focuses on deep learning technologies and neural networks.



Consumer goods - Barbie



- Using natural language processing, machine learning and advanced analytics, Hello Barbie listens and responds to a child.
- A microphone on Barbie's necklace can record and transmit to the servers at ToyTalk.
- There, the recording is analyzed to determine the appropriate response from 8,000 lines of dialogue.
- Servers transmit the correct response back to Barbie in under a second so she can respond to the child.

Some more..

HR Analytics



Facial Recognition



Recommendations

Frequently bought together



Sponsored products related to this item (What's this?)



Roles in the Industry

Data Science Roles



Data Science Developers



Big Data Specialists



Data Engineers



Data Scientist



Researchers



Analysts

DS Infrastructure Engineers

- Data Science Engineers are hard-core techies who deal with Data Science Infrastructure, ie., hardware, software applications and other aspects to get the back end of Data Science up and running.
- They set up the entire IT infrastructure from servers, networks to processes, also manage it such as infrastructure monitoring, application management, database administration etc.,



Data Science Developers



- Data Science Developers are the ones, who code the models and applications through programming in R Language, Python etc., They are versatile developers, who have good knowledge on math & statistics, machine learning algorithms and related concepts.
- As this domain of data science development is evolving rapidly, these developers are expected to keep themselves updates with all latest technological advances from development perspective, so that they can use the right platform to achieve their goal in effective manner.

Machine Learning Specialist

- These are the professional with deep knowledge in computer science and mathematics.
- They engage in machine learning and deep learning heavily.
- They create predictive and prescriptive models based on the machine learning algorithms, such as random forest, Artificial Neural Network, K-NN etc.,
- They are masters in all kinds of data mining techniques pruning, regularization etc., helping to create a robust data science models, which can be used in creating great business insights.



Big Data Specialists

- Big Data Specialists focus on designing model for of Big Data Processing as one of the building block of Data Science workflow
- Their work involve architecture of Data gathering including streaming and snapshots, storing and processing in effective and efficiently manner using Big Data Technologies



Researchers

- Researchers are domain experts. These professionals have more expertise in Data analysis, Data Science related Statistics along with significant expertise in specific domain such as HR, Marketing, Fraud Analytics, Health Care, Finance etc.,
- They have less emphasis and knowledge in backend part such as IT infrastructure, coding, computer science etc.,



Data Science Analyst

- These professionals are focussed on day-to-day analysis of data, including website analytics, retrieving data from various data sources and creating data visualizations.
- They work closely with business person. Their role is to provide the reports from data analysis with appropriate visualizations in an easy to understand format. There by enabling the decision makers to gain valuable business insights



Business Person

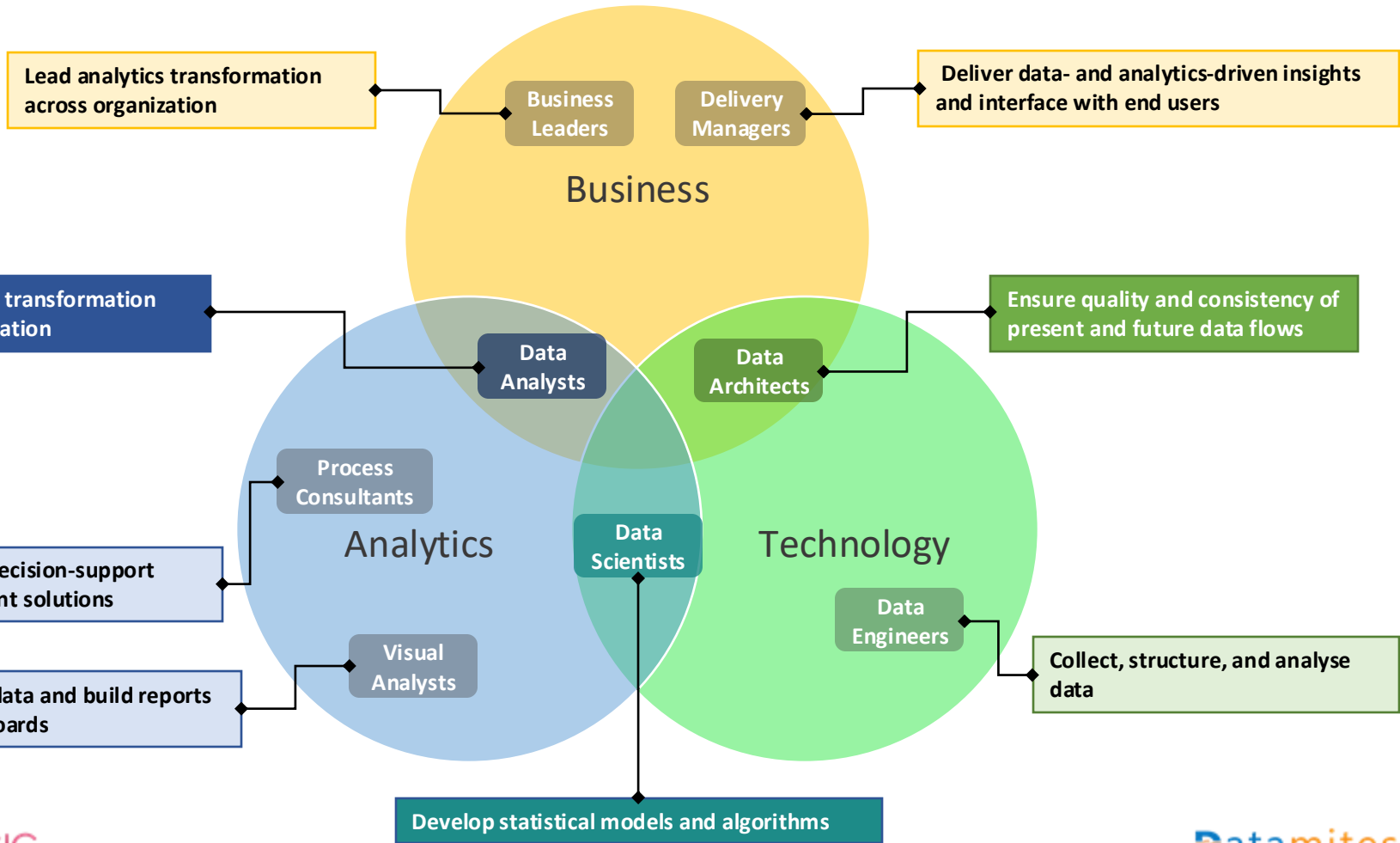
- This role is the one, who manages the entire Data Science project. We can call him as sponsor. This role is predominantly business focussed and, formulates the problem statements for which, Data science project needs to find the answers. This role also helps in understanding, interpreting the intermediate results of the data science projects and drive the project to final solution.
- Though this role is primarily business focussed, he/she must also speak Data in order to able to perform the job well.



Data Scientist

- These are the professional who is able to perform every aspect of Data Science, sometime called as full-stack Data science professionals. Well, these people are rare as mastering entire Data Science roles is difficult, if not impossible.
- If a company manages to hire full-stack Data Scientist, there would be tremendous progress in transforming the business in gain significant competitive advantage by finding solutions to important business questions.





Installations

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